

Ground states of finite PEPS from a generating function: belief propagation first, then matrix-product BP environments

TensorNetworkQuantumSimulator, FixesV2DMRG

September 2026

The idea in one paragraph

We want the ground state of a local Hamiltonian $H = \sum_e h_e$ in a finite PEPS $|\psi\rangle$ on an open square lattice. The usual variational route contracts the indefinite network $\langle\psi|H|\psi\rangle$. We never do that. Instead we embed H in a *positive* norm network through the generating operator

$$G(\lambda) = \prod_e (1 + \lambda h_e), \quad T(\lambda) = \langle\psi|G(\lambda)|\psi\rangle = \langle\psi|\psi\rangle + \lambda \langle\psi|H|\psi\rangle + O(\lambda^2), \quad (1)$$

so that the energy is a λ -derivative of a partition function, $E = \partial_\lambda \ln T(\lambda)|_{\lambda=0}$, and every environment we ever build is the positive double-layer environment of a norm-like network. Each edge factor $1 + \lambda h_e$ is written with one auxiliary index of dimension $r + 1$: slot 0 carries the identity (the norm sector), slots $1 \dots r$ carry the operator-Schmidt factors $A_a \otimes B_a$ of h_e . The whole approximate-contraction machinery of the package (BP at $\chi = 1$, finite CTMRG at $\chi > 1$) then applies to $T(\lambda)$ unchanged. With graded (symmetric) tensors the auxiliary index is a direct sum of the trivial sector and the Schmidt bond, so $\mathbb{Z}_2$ and $U(1)$ come for free.

Energy and gradient without a Hessian

Write $Z_\chi(\lambda)$ for the approximate contraction of $T(\lambda)$ at environment dimension χ (Bethe free energy at $\chi = 1$, the CVM/CTM free energy F at $\chi > 1$) and $X(\lambda)$ for its converged environments. Two facts do all the work.

- **Envelope theorem.** Because X is a stationary point of the free energy, $\partial_\lambda F(T(\lambda), X(\lambda)) = \partial_\lambda F(T(\lambda), X)|_{X \text{ fixed}}$ to first order. The energy is therefore a sum of local ratios over the vertex regions with the operator replaced by $\partial_\lambda G$, no response of the environment needed. In practice we record the implicit central difference $E = [F(+\lambda) - F(-\lambda)]/2\lambda$ with the environments re-converged at $\pm\lambda$ ($\lambda = 10^{-7}$): under the truncating CTM projector this carries the projector response and is 10^2 – 10^3 times more accurate than the fixed-environment formula, at no extra cost beyond the two environments we need anyway.
- **Effective operators by finite difference.** At a vertex v the environment of $T(\lambda)$ is a ring $R_\lambda(v)$ around the site. The norm and Hamiltonian effective operators on the tensor t_v are

$$N_{\text{eff}} = \frac{1}{2} [R_{+\lambda} \cdot G_0 + R_{-\lambda} \cdot G_0], \quad H_{\text{eff}} = \frac{R_{+\lambda} \cdot G(+\lambda) - R_{-\lambda} \cdot G(-\lambda)}{2\lambda}, \quad (2)$$

dense $\dim(t_v) \times \dim(t_v)$ matrices (162×162 at $D = 3$). H_{eff} carries an arbitrary multiple of N_{eff} (the λ -derivative of the block rescaling), which shifts the local eigenvalue but not the eigenvector and cancels exactly in the gradient

$$\frac{\partial E}{\partial \bar{t}_v} = \frac{2(H_{\text{eff}} - \varepsilon N_{\text{eff}}) t_v}{t_v^\dagger N_{\text{eff}} t_v}, \quad \varepsilon = \frac{t_v^\dagger H_{\text{eff}} t_v}{t_v^\dagger N_{\text{eff}} t_v}. \quad (3)$$

Nothing is inverted; no Hessian appears. Both operators come from the same three environments, so *one* environment set gives the exact gradient at *every* vertex.

The two stages

Stage 1: $\chi = 1$ (belief propagation). The environments are BP messages of $T(0)$; their linear response to λ is a Gauss–Seidel sweep on the one-insertion sector (exact on a tree). A one-site sweep solves the local generalised eigenproblem $H_{\text{eff}} t = \varepsilon N_{\text{eff}} t$ matrix-free (whitened Lanczos) at each vertex, with a warm-started response, an acceptance test on the Bethe energy and a damped retry. This costs seconds per sweep and converges in 2–3 sweeps. It minimises the *Bethe* energy, which is itself 10^{-2} per site off; what it produces is a state whose true energy is already within 10^{-5} – 10^{-6} per site of exact on a 5×5 (Table 1).

Stage 2: $\chi > 1$ (matrix-product BP / finite CTMRG). The environments are the $4C + 4T$ rings of a position-resolved finite CTMRG at interface dimension χ (projector :cut). Because one environment set yields the whole gradient, the natural optimiser is global: L-BFGS over all tensors at once, with the block variable metric $H_0 = \gamma N_{\text{eff}}^{-1}$ on the whitened subspace (directions with N_{eff} weight below 10^{-6} of the maximum are dropped; γ is the Barzilai–Borwein scalar). The first step, and any fallback, is the Jacobi direction $t_v^* - t_v$ towards the local one-site optimum. Every step is an Armijo backtracking line search on $E = [F(+\lambda) - F(-\lambda)]/2\lambda$, so no accepted step raises the energy, and tensors are renormalised after each step. From the BP state the line search accepts the unit L-BFGS step at essentially every iteration.

Cost. An environment set is three CTM convergences ($\lambda = 0, \pm\lambda$). The $\lambda = 0$ one is built on the plain norm network and padded onto the generating network (the $a > 0$ slots carry zero weight at $\lambda = 0$), so it costs a fraction of the other two; the $\pm\lambda$ pair, which needs the auxiliary leg, is now 94% of a set. On the 5×5 at $D = 3$, $\chi = 48$ a set is 50 s, so an L-BFGS iteration is about a minute against 40 minutes for a one-site sweep with per-vertex refreshes of the same environments.

What it gives

system	stage	$D = 3$	$D = 4$
5×5 TFIM, $g = 3$, error per site vs exact	simple update (start)	2.5×10^{-5}	6.5×10^{-5}
	stage 1 (BP, $\chi = 1$)	1.9×10^{-5}	1.3×10^{-6}
	stage 2 (CTM L-BFGS, $\chi = 48$)	8.2×10^{-7}	1.6×10^{-7}
5×5 , $g = 3.04438$, $D = 3$	Yantao Wu, VMC + SR	7.8×10^{-7}	
	stage 2 started from that state	6.7×10^{-7} (floor)	
6×6 , $g = 3$, $D = 4$	E/site : SU, BP, CTM	$-3.15477533, -3.15477644, -3.15477967$	

Table 1: Energy density errors against exact diagonalisation (matrix-free Lanczos on 2^{25} states) on the 5×5 transverse-field Ising model $H = -\sum ZZ - g\sum X$, and energies on the 6×6 where no exact reference exists (boundary MPS at $\chi = 32$ agrees with the optimiser’s own estimate to 10^{-9}). The $D = 4$ stage-2 row is three iterations; the $D = 3$ row 22 iterations, still descending. The optimiser lowers an independently VMC-optimised state and then finds no downhill direction, which locates the $D = 3$ floor at this χ .

What limits it, and what is next

The cost is the $\pm\lambda$ environment pair; `:cycle`, the stationary projector whose gradients are far more accurate, is not yet usable because its refresh after a state change runs to the iteration cap. Larger bond dimension needs longer runs, not new ideas: at $D = 4$ on a 6×6 one environment set is eight minutes. The full derivation, every measured number and the falsified alternatives are in `docs/dmrg.md`; the code is `dmrg( $\psi$ , H; alg = "bp")` for stage 1 and `dmrg( $\psi$ , H; alg = "ctmrg_lbfgs", maxdim)` for stage 2 in `src/dmrg.jl`.